

Opinion Response-Pattern Network Analysis

DPCN Assignment 1 Report

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Abstract

This report contains analysis of an opinion response-pattern network from a 60-item survey covering Technology, Education, Social/Ethics, and Environment. Survey responses were converted to ordinal Likert scores and pairwise Spearman rank correlations were used to quantify similarity between question-response patterns. A thresholded weighted network was then constructed, with $T = 0.40$ selected as the working threshold. The resulting graph contains a 40-node largest connected component (LCC) with 90 edges. Weighted Louvain community detection identifies six thematic communities, while centrality analysis highlights structurally important questions and bridges between parts of the network. Additional analysis compares structural centrality with direct response agreement and examines the relationship between the original survey categories and the detected communities.

1. GitHub Link to Code

https://github.com/Sylvester4582/DPCN__LUCH_A1

The repository contains the notebook used for data preprocessing, network construction, threshold analysis, community detection, centrality computation, and visualization.

2. Dataset Documentation

The dataset analyzed in this work is the survey response file `Survey_Results_UC.csv`. The survey contains 60 opinion statements divided equally among four categories:

- **Technology:** T01–T15
- **Education:** E01–E15
- **Social/Ethics:** S01–S15
- **Environment:** V01–V15

The original responses used the five-point Likert scale *Strongly Disagree*, *Disagree*, *Neutral*, *Agree*, *Strongly Agree*, together with *No Comments*. The categorical responses were encoded numerically as

$$\text{StronglyDisagree} = 1, \quad \text{Disagree} = 2, \quad \text{Neutral} = 3,$$

$$\text{Agree} = 4, \quad \text{StronglyAgree} = 5.$$

Responses marked *No Comments*, as well as blank cells, were treated as missing values and represented by NaN. Question columns were renamed using compact identifiers such as T01 and E10, while the original question text was retained separately as metadata for interpretation.

Since Likert responses are ordinal, Spearman rank correlation was used rather than Pearson correlation. For two questions i and j , the resulting coefficient ρ_{ij} measures the similarity in the ordering of respondents' answers across the two questions. Pairwise correlations were computed using the available non-missing responses.

3. Pipeline Followed

The complete network construction pipeline is summarized below.

1. **Data loading and normalization:** The survey data were loaded, question columns were mapped to short IDs, and the categorical Likert responses were converted to ordinal values from 1 to 5.
2. **Correlation construction:** A 60×60 Spearman correlation matrix was computed across all survey items.
3. **Network construction and thresholding:** A weighted edge between questions i and j was retained only when

$$\rho_{ij} > T.$$

Thus, the network captures positive response-pattern similarity above the selected threshold.

4. **Threshold sensitivity analysis:** Thresholds from 0.30 to 0.60 were tested. The number of edges, density, connected components, LCC size, isolates, and clustering were compared.
5. **Working threshold selection:** $T = 0.40$ was selected as the working threshold because it provides a substantially sparser network than lower thresholds while still retaining a sizeable 40-node connected component.

6. **Graph decomposition:** At $T = 0.40$, the thresholded graph contains 91 edges in total. The largest connected component contains 40 nodes and 90 edges. The remaining nodes consist of an isolated pair, E05–E06, connected by one edge, and 18 isolated nodes.
7. **Community detection:** Weighted Louvain community detection was applied to the 40-node LCC using seed=42. This produced six communities with modularity

$$Q = 0.3676.$$

8. **Centrality analysis:** Degree, weighted degree (strength), betweenness, closeness, eigenvector centrality, and weighted local clustering coefficients were computed.
9. **Response-level analysis:** Mean response, response variance, agreement rate, disagreement rate, and Shannon entropy were calculated to distinguish direct response agreement from structural importance in the network.
10. **Community-domain comparison:** The composition of each Louvain community was compared with the original four survey categories to determine whether the detected structure was simply domain-based or reflected cross-domain relationships.

4. Analysis and Visualizations

4.1. Threshold Sensitivity Analysis

The threshold determines how strongly correlated two questions must be before they are connected. Table 1 shows that increasing the threshold rapidly removes edges and breaks the graph into smaller components.

Table 1. Network topology across correlation thresholds.

T	Edges	Density	Avg. Degree	Components	LCC	Isolates	Clustering
0.30	322	0.1819	10.7333	8	53	7	0.5508
0.35	188	0.1062	6.2667	14	47	13	0.4410
0.40	91	0.0514	3.0333	20	40	18	0.2826
0.45	43	0.0243	1.4333	32	24	26	0.2558
0.50	13	0.0073	0.4333	48	7	42	0.3000
0.55	7	0.0040	0.2333	53	5	50	0.0000
0.60	2	0.0011	0.0667	58	3	57	0.0000

At $T = 0.30$, the network is considerably denser and retains a 53-node LCC, whereas at $T = 0.45$ the LCC falls to 24 nodes, with 26 nodes becoming isolates. The selected value $T = 0.40$ therefore provides a useful compromise between retaining a sufficiently large connected structure and removing weaker correlations.

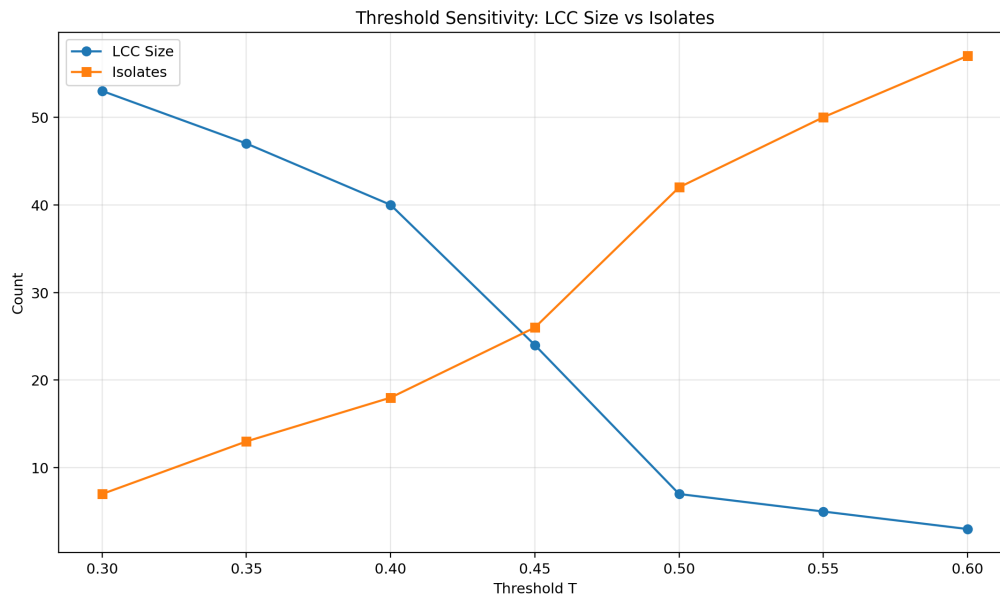


Figure 1. Threshold sensitivity showing the reduction in LCC size and increase in isolates as the correlation threshold increases.

4.2. Graph Decomposition and Global Topology

For the selected working threshold $T = 0.40$, the complete thresholded graph contains 91 edges. Of these, 90 belong to the LCC, while one edge connects the pair E05 and E06. The remaining 18 questions are isolates.

The LCC contains 40 nodes and 90 edges. Its average degree is 4.5 and its density is 0.1154, indicating a relatively sparse graph. Nevertheless, the network remains connected within the LCC, allowing paths between all included questions.

Table 2. Global topology of the 40-node largest connected component.

Metric	Value
LCC Nodes	40
LCC Edges	90
Average Degree	4.5000
Graph Density	0.1154
Diameter	8
Average Shortest Path Length	2.9564
Global Transitivity	0.2826
Mean Local Clustering	0.1468

The diameter of 8 indicates that the most distant pair of questions requires eight network steps to connect, while the average shortest path is approximately 2.96. Global transitivity is 0.2826, showing the presence of local triangular structure without the graph becoming highly clustered.

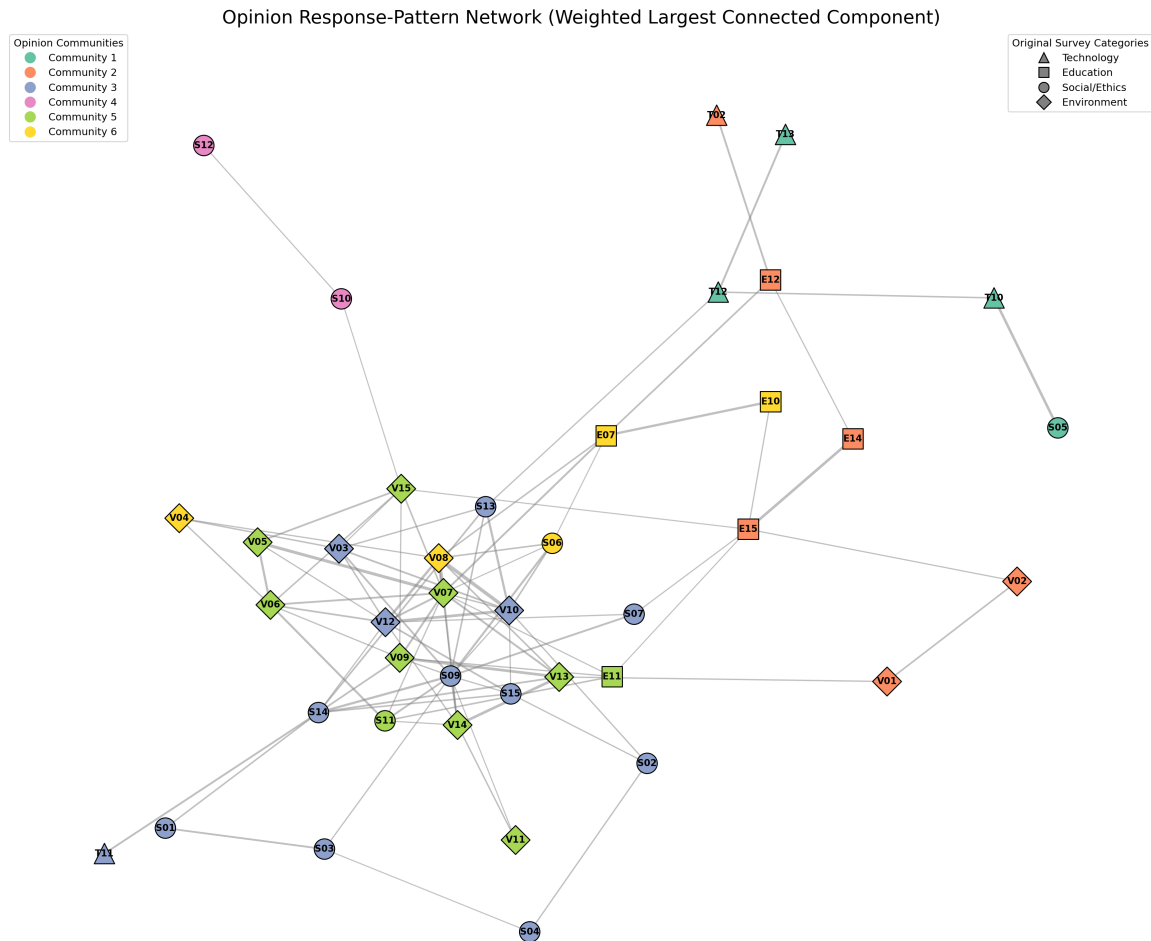


Figure 2. Weighted opinion response-pattern network for the LCC at $T = 0.40$. Node shape denotes the original survey category and node color denotes Louvain community. Edge width reflects correlation strength.

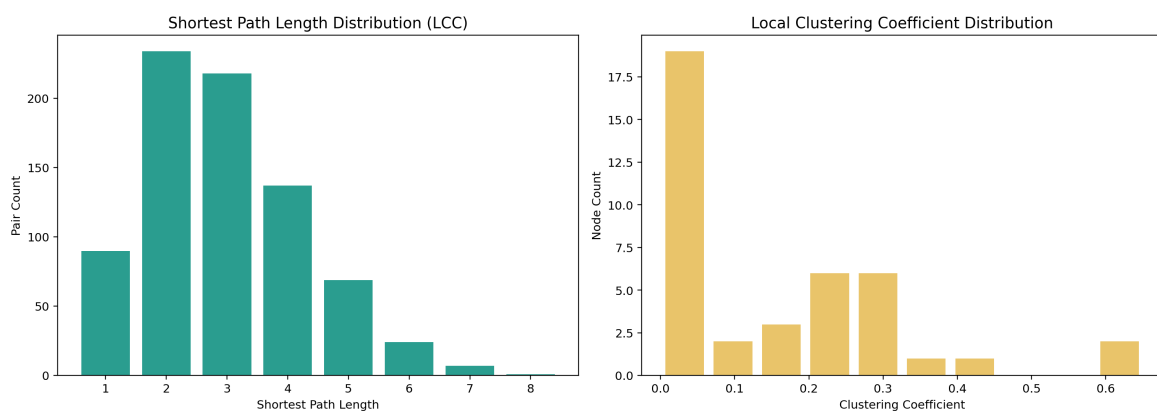


Figure 3. Shortest-path length and local clustering coefficient distributions in the LCC.

4.3. Centrality Analysis

Different centrality measures reveal different structural roles. Betweenness centrality identifies questions that lie on many shortest paths and can therefore connect otherwise weakly connected

regions. Eigenvector centrality identifies nodes that are connected to other structurally important nodes.

The strongest structural bridges are shown below.

Table 3. Top structural bridges by betweenness centrality.

Question	Category	Betweenness	Strength
S09	Social/Ethics	0.2013	5.4405
S13	Social/Ethics	0.1919	2.2959
T12	Technology	0.1484	1.3600
V15	Environment	0.1455	3.0072
V07	Environment	0.1427	6.2802

S09 has the largest betweenness centrality at approximately 0.2013, followed by S13 at 0.1919. These questions occupy strategically important positions in the network and connect regions that would otherwise be less directly connected.

Eigenvector centrality highlights a different structural property.

Table 4. Top structurally central questions by eigenvector centrality.

Question	Category	Eigenvector	Closeness
V07	Environment	0.4041	0.4815
V12	Environment	0.3214	0.4432
V08	Environment	0.3018	0.4382
V10	Environment	0.2967	0.4432
S09	Social/Ethics	0.2667	0.4815

V07 has the highest eigenvector centrality (0.4041), followed by V12, V08, and V10. The strong representation of Environmental questions among the most central nodes indicates that sustainability-related opinions occupy an important position in the core of the LCC.

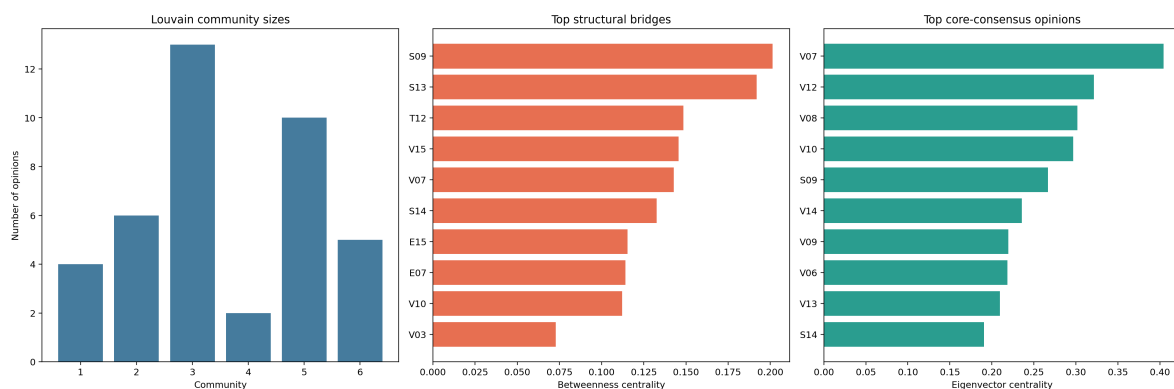


Figure 4. Louvain community sizes, top betweenness nodes, and top eigenvector-centrality nodes.

4.4. Structural Centrality versus Direct Response Agreement

Network centrality and direct survey agreement represent different properties. A question may be strongly connected to many other important questions without necessarily receiving the highest level of agreement from respondents.

The response-level analysis confirms this distinction. V07, for example, has the highest eigenvector centrality, while E15 has the highest direct agreement rate.

Table 5. Questions with the highest direct agreement rates.

Question	Category	Agreement	Mean	Entropy
E15	Education	88.54%	4.7126	0.9459
S11	Social/Ethics	87.50%	4.5517	1.1737
V13	Environment	86.46%	4.6706	1.0158
E10	Education	85.42%	4.5287	1.2422
S05	Social/Ethics	85.42%	4.6322	1.1081

The low entropy values of these questions are consistent with concentrated responses. In contrast, eigenvector centrality is a network-level property and should not be interpreted as a direct measure of agreement.

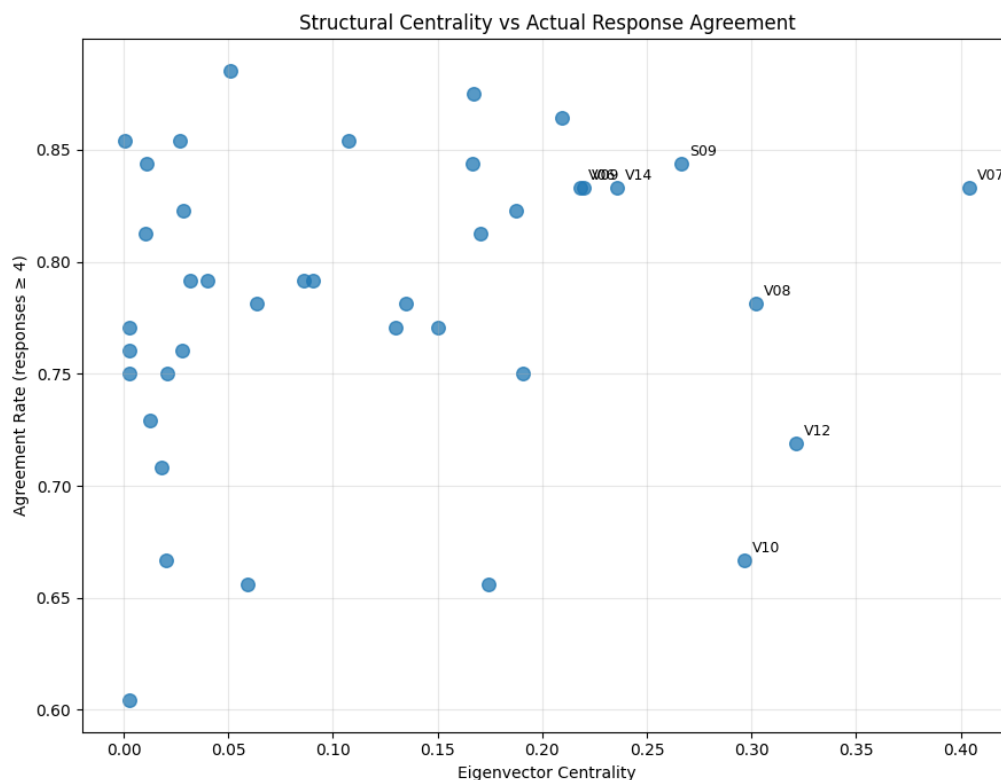


Figure 5. Relationship between eigenvector centrality and direct agreement rate.

4.5. Domain Interaction

The original questionnaire contains four predefined domains, but the response-pattern network is not restricted to within-domain relationships. Among the 90 edges in the LCC, 56 are within-domain

and 34 are cross-domain.

Table 6. Within-domain and cross-domain edge distribution.

Relation	Edges	Mean Correlation	Share
Within-Domain	56	0.4704	62.22%
Cross-Domain	34	0.4427	37.78%

The domain mixing matrix further shows that Social/Ethics and Environment have particularly strong interaction, with 21 cross-domain edges between them. Education also interacts with Environment through six edges, while Technology has comparatively fewer links to Environment.

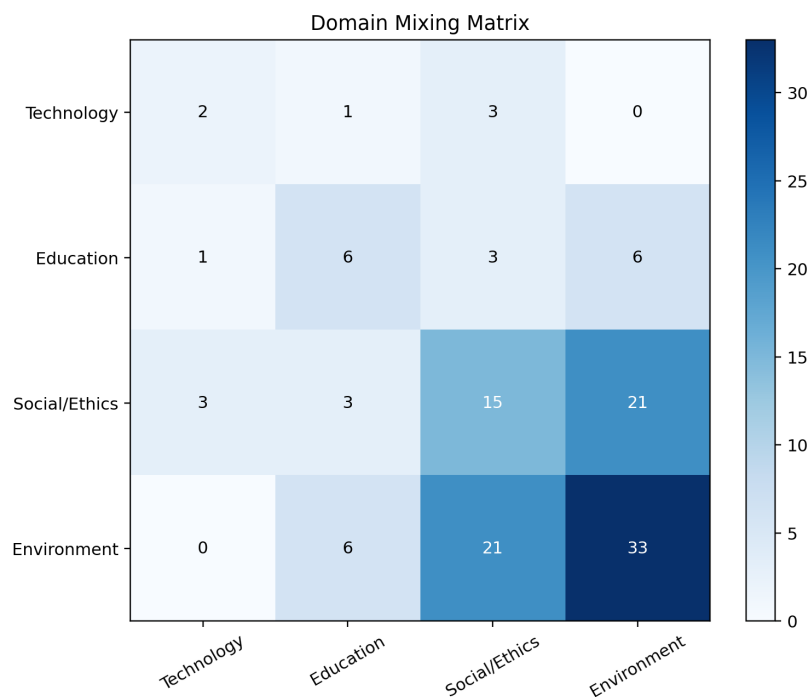


Figure 6. Domain mixing matrix showing the number of relationships between the four original survey categories.

4.6. Louvain Communities and Thematic Structure

Weighted Louvain community detection using seed=42 produced six communities with modularity $Q = 0.3676$. The six groups can be interpreted according to the questions they contain:

- **Community 1 (4 items):** *Data Privacy, Cybersecurity & AI Governance*. Contains T10, T12, T13, and S05, forming a cross-cutting Technology–Ethics group.
- **Community 2 (6 items):** *Institutional Adaptation, AI in Education & Renewable Energy Policy*. Contains E12, E14, E15, T02, V01, and V02.
- **Community 3 (13 items):** *Social Responsibility, Corporate Accountability & Everyday Sustainability*. This is the largest detected community and combines Social/Ethics questions with several Environmental items.
- **Community 4 (2 items):** *Evidence-Based Policy & Youth Civic Engagement*. Contains S10 and S12.

- **Community 5 (10 items):** *Ecological Conservation, Global Treaties & Circular Economy*. It is predominantly Environmental, containing eight Environment questions along with E11 and S11.
- **Community 6 (5 items):** *Higher Education Innovation & Sustainable Campus Operations*. Contains E07, E10, V04, V08, and S06.

Table 7. Composition of Louvain communities by original survey category.

Community	Education	Environment	Social/Ethics	Technology	Size
1	0	0	1	3	4
2	3	2	0	1	6
3	0	3	9	1	13
4	0	0	2	0	2
5	1	8	1	0	10
6	2	2	1	0	5

The normalized mutual information between the original survey category and the Louvain community assignment is

$$NMI = 0.3589.$$

This indicates that the detected communities have some relationship with the original survey domains, but the community structure is not equivalent to simply separating the questions into Technology, Education, Social/Ethics, and Environment.

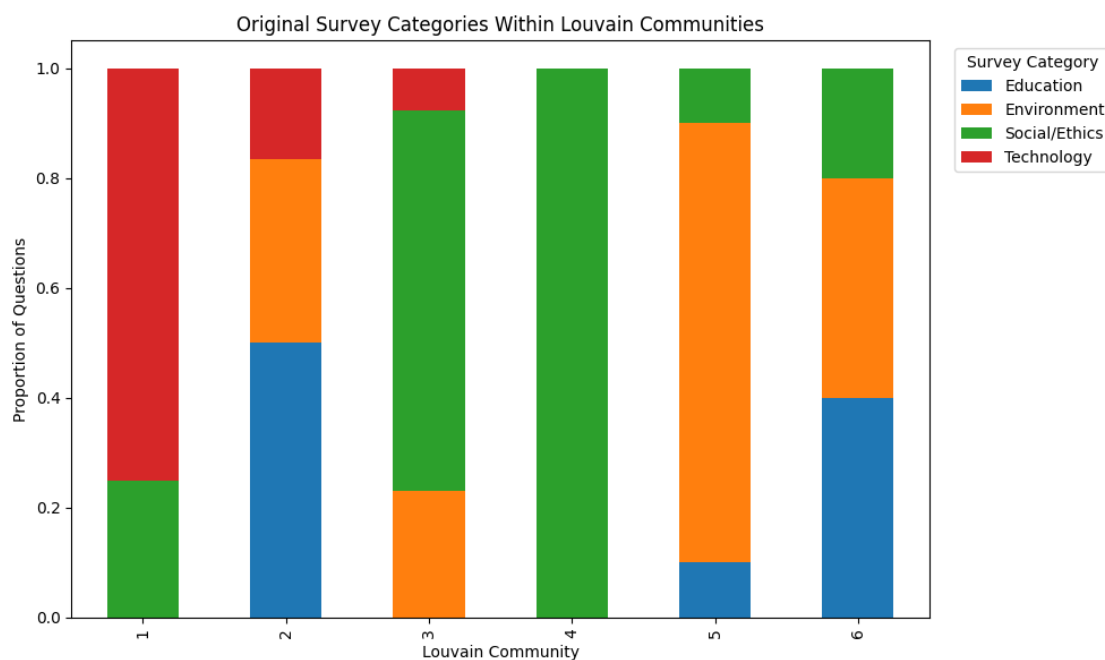


Figure 7. Original survey-category composition within the six Louvain communities.

4.7. Community Detection Robustness

The Louvain algorithm was run using six different random seeds: 0, 1, 2, 3, 4, and 42. The number of detected communities varied between five and six, while modularity ranged from 0.3317 to 0.3741.

To quantify partition similarity, Adjusted Rand Index (ARI) and Normalized Mutual Information (NMI) were computed across the different runs. The mean pairwise values were

$$\overline{ARI} = 0.4795, \quad \overline{NMI} = 0.6444.$$

The selected seed 42 produced a partition particularly similar to the seed-1 partition, with

$$ARI = 0.9232, \quad NMI = 0.9569.$$

At the same time, the variation across other seeds shows that the precise community membership should not be treated as a unique or perfectly invariant partition. The six communities used for substantive interpretation therefore represent the structure obtained at the selected working seed rather than an absolute partition of the network.

5. Results and Discussion

The response-pattern network reveals several important characteristics of the survey.

First, the choice of threshold has a substantial effect on network structure. At $T = 0.30$, the graph is considerably denser and retains a 53-node LCC, whereas at $T = 0.45$ the LCC falls to only 24 nodes. The selected working threshold of $T = 0.40$ produces a 40-node LCC with 90 edges, providing a sufficiently connected structure for examining relationships while excluding weaker correlations.

Second, the network contains meaningful structural hubs. S09 has the highest betweenness centrality (0.2013), indicating that it lies on a large number of shortest paths and connects different portions of the graph. S13 is the second-highest bridge, while T12, V15, and V07 also occupy relatively important bridging positions. The largest eigenvector centrality is observed for V07 (0.4041), followed by V12, V08, and V10. Thus, environmental questions are strongly represented among the most structurally central opinions.

Third, structural importance should be distinguished from direct response agreement. E15 has the highest observed agreement rate at 88.54%, while V07 has the highest eigenvector centrality. Therefore, the statement that a question is central in the network does not automatically imply that it has the highest level of agreement among respondents. The response-level analysis provides a complementary view of the class opinions.

Fourth, the graph exhibits substantial cross-domain interaction. Of the 90 LCC edges, 34 (37.78%) are cross-domain relationships. The strongest interaction in the domain mixing matrix is between Social/Ethics and Environment, with 21 edges. This supports the interpretation that opinions concerning social responsibility, ethics, environmental stewardship, and institutional behavior are related across the original categories rather than existing as completely separate domains.

The community analysis reinforces this observation. Community 5 is strongly Environment-oriented, while Community 1 primarily contains Technology questions with one Social/Ethics

item. Other groups mix Education, Environment, and Social/Ethics questions. The category-community NMI of 0.3589 indicates partial alignment with the original domains, but not a one-to-one correspondence. The detected communities therefore capture response-pattern relationships that cut across the questionnaire's original organization.

The isolated components also provide a useful observation. Questions E05 and E06 form a separate two-node component, while 18 questions are isolated at the selected threshold. The pair represents a localized Education-related relationship that does not reach the 0.40 correlation threshold with the LCC. Similarly, an isolated question means that no other question has a positive Spearman correlation above 0.40 with it; isolation alone does not establish that the underlying opinion is polarized.

Finally, the Louvain robustness analysis indicates that the broad community structure is meaningful but that the exact partition depends partly on algorithm initialization. Across the tested seeds, five or six communities were obtained. This suggests that the semantic community labels should be interpreted as an informative description of the selected partition rather than as uniquely determined clusters.

Overall, the network shows that the survey responses are not organized purely according to the four predefined subject areas. Environmental and Social/Ethics questions occupy important central and bridging positions, while several communities combine questions from multiple domains. The network therefore captures higher-level associations in how respondents answered related social, institutional, technological, and environmental statements.

6. Individual Contribution

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Data preprocessing, coding, network construction, correlation computation, threshold sensitivity analysis, and graph decomposition.

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Weighted Louvain community detection, community robustness analysis, centrality metrics, and structural network analysis.

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Visualization, domain interaction analysis, response-level analysis, substantive interpretation, report drafting, and final formatting.